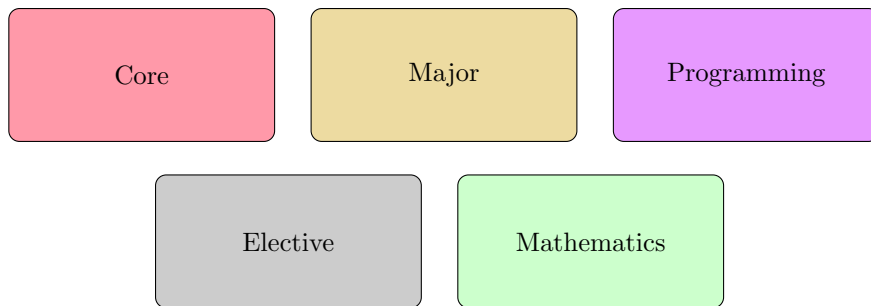


BMath handbook

August 12, 2026

1 Introduction

In this document we present suggested pathways for completing the Bachelor of Mathematics and Bachelor of Mathematics (Advanced) programs. Depending on your entry time to the university and your existing knowledge, we provide a recommendation for which courses you should enrol in and when as a diagram. We have coloured the courses as



Where we have:

- **Core:** courses that must be undertaken for the program
- **Major:** courses that contribute towards a major
- **Programming:** your choice of one of ENGG1003, INFT1004, or SENG1110
- **Elective:** your choice of any course offered by the university
- **Mathematics (only applicable for the BMath Advanced):** a choice between MATH1110 and SCIE3500 depending on your existing knowledge

2 Bachelor of Mathematics

Majors

There are three (3) possible majors for the Bachelor of Mathematics:

- **Pure and Applied Mathematics:** For those who wish to immerse themselves in fundamental mathematics such as algebra and number theory, and to model real world scenarios by constructing and solving ordinary and partial differential equations by hand and on a computer.
- **Statistics:** For those that want to deal with the uncertainty of life, modelling real world systems with different data types, optimising models to better reflect data, and predicting the future with time series forecasting.
- **Studies in Mathematics and Statistics:** This major is for those who want to pick and choose courses from the other two majors.

Entry point

Many students will enter university after having completed their NSW HSC in the last two years. If you have successfully completed Advanced Mathematics—with a band score of 5 or higher—or Extension 1 or 2 Mathematics then **MATH1110** is the recommended entry point.

If you have a comparable background, but you haven't completed high school in the last two years or not in NSW, then you must pass the [Math Placement Test](#) to enrol into MATH1110.

Alternatively, the course **MATH1002** prepares those who don't meet the requirements for MATH1110. If you don't meet the requirements for MATH1110, you can start with MATH1002 in your first semester. You can then take MATH1110 in the following semester or during the Midyear Session, followed by MATH1120 in the next available semester.

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2.1 Major: Pure and Applied Mathematics

MATH3000s course options

As part of this major you only need to choose four of the five MATH3000s level courses. The MATH3000s courses can be broadly split between pure and applied, use the following table to guide your choice of third year courses. All five MATH3000s level courses can be taken by using a generic elective slot.

	Skews pure	Skews applied
Semester 1	MATH3120 - Algebra MATH3205 - Applied Fourier Analysis	MATH3820 - Numerical Methods
Semester 2	MATH3170 - Number Theory	MATH3700 - Partial Differential Equations

2.1.1 Semester 1 entry; starting with MATH1110

2027	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	Elective
	Semester 2	STAT2020 Predictive Analytics	MATH2242 Complex Analysis	Elective	Elective
2029	Semester 1	Semester 1 MATH3000s course	Semester 1 MATH3000s course	Elective or Semester 1 MATH3000s course	Elective
	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	Semester 2 MATH3000s course	Semester 2 MATH3000s course or Elective	Elective

Note: MATH3700 can be taken in 2028 semester 2.

2.1.2 Semester 1 entry; starting with MATH1002

2027	Semester 1	MATH1002 Foundational Studies in Mathematics	STAT1100 Data Communication and Modelling	Programming	Elective
	Mid-year	MATH1110 Mathematics for Engineering, Science and Technology 1			
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	Elective
	Semester 2	STAT2020 Predictive Analytics	MATH2242 Complex Analysis	Elective	Elective
2029	Semester 1	Semester 1 MATH3000s course	Semester 1 MATH3000s course	Elective or Semester 1 MATH3000s course	Elective
	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	Semester 2 MATH3000s course	Semester 2 MATH3000s course or Elective	

Note: MATH3700 can be taken in 2028 semester 2.

2.1.3 Semester 1 entry; starting with MATH1002 (no midyear)

2027	Semester 1	MATH1002 Foundational Studies in Mathematics	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1300 Fundamentals of Statistics	Elective	Elective
2028	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	Elective	Elective	Elective
	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	MATH2370 Real Analysis
2029	Semester 1	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	MATH3205 Applied Fourier Analysis	MATH3820 Numerical Methods
	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	MATH2242 Complex Analysis	MATH3170 Number Theory	MATH3700 Partial Differen- tial Equations

It is strongly recommended that you take MATH1110 during the 2027 Mid-year term, so that you can then continue with the previous pathway from 2027 Semester 2 onwards.

2.1.4 Semester 2 entry; starting with MATH1110

2027	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1300 Fundamentals of Statistics	Programming	Elective
	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	Elective	Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	MATH2370 Real Analysis
	Semester 1	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	Elective	Elective
2029	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	MATH2242 Complex Analysis	Semester 2 MATH3000s course	Semester 2 MATH3000s course or Elective
	Semester 1	Semester 1 MATH3000s course	Semester 1 MATH3000s course	Elective or Semester 1 MATH3000s course	Elective

Note: MATH3820 can be taken in 2029 semester 1.

2.1.5 Semester 2 entry; starting with MATH1002

2027	Semester 2	MATH1002 Foundational Studies in Mathematics	Programming	Elective	Elective
	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Elective	Elective
2028	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	Elective
2029	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT2020 Predictive Analytics	MATH2242 Complex Analysis	Semester 2 MATH3000s course
	Semester 1	MATH3120 Algebra	MATH3205 Applied Fourier Analysis	MATH3820 Numerical Methods	Elective
2030	Semester 1				

2.2 Major: Statistics

2.2.1 Semester 1 entry; starting with MATH1110

2027	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	STAT2000 Applied Statistics and Research Methods	Elective
	Semester 2	STAT2020 Predictive Analytics	STAT2300 Statistical Inference	Elective	Elective
2029	Semester 1	STAT3030 Generalised Lin- ear Models	STAT3800 Deterministic and Stochastic Optimisation	Elective	Elective
	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT3040 Time Series Forecasting	STAT3100 Systems Think- ing for an Inte- grated Workforce	Elective

2.2.2 Semester 1 entry; starting with MATH1002

2027	Semester 1	MATH1002 Foundational Studies in Mathematics	STAT1100 Data Communication and Modelling	Programming	Elective
	Mid-year	MATH1110 Mathematics for Engineering, Science and Technology 1			
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	STAT2000 Applied Statistics and Research Methods	Elective
	Semester 2	STAT2020 Predictive Analytics	STAT2300 Statistical Inference	Elective	Elective
2029	Semester 1	STAT3030 Generalised Lin- ear Models	STAT3800 Deterministic and Stochastic Optimisation	Elective	Elective
	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT3040 Time Series Forecasting	STAT3100 Systems Think- ing for an Inte- grated Workforce	

2.2.3 Semester 1 entry; starting with MATH1002 (no midyear)

2027	Semester 1	MATH1002 Foundational Studies in Mathematics	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1300 Fundamentals of Statistics	Elective	Elective
2028	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT2000 Applied Statistics and Research Methods	Elective	Elective
	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	STAT2300 Statistical Inference
2029	Semester 1	MATH2360 Linear Algebra	STAT3030 Generalised Lin- ear Models	STAT3800 Deterministic and Stochastic Optimisation	Elective
	Semester 2	MATH2370 Real Analysis	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT3040 Time Series Forecasting	STAT3100 Systems Think- ing for an Inte- grated Workforce

It is strongly recommended that you take MATH1110 during the 2027 Mid-year term, so that you can then continue with the previous pathway from 2027 Semester 2 onwards.

2.2.4 Semester 2 entry; starting with MATH1110

2027	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1300 Fundamentals of Statistics	Programming	Elective
	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	STAT2000 Applied Statistics and Research Methods	Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	STAT2300 Statistical Inference
	Semester 1	MATH2360 Linear Algebra	STAT3030 Generalised Lin- ear Models	STAT3800 Deterministic and Stochastic Optimisation	Elective
2029	Semester 2	MATH2370 Real Analysis	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT3040 Time Series Forecasting	STAT3100 Systems Think- ing for an Inte- grated Workforce
	Semester 1	Elective	Elective	Elective	Elective
2030	Semester 1	Elective	Elective	Elective	Elective

Note: STAT3030 and STAT3800 can be delayed until 2030 semester 1.

2.2.5 Semester 2 entry; starting with MATH1002**WARNING:** It is impossible to complete this plan in 3 years.

2027	Semester 2	MATH1002 Foundational Studies in Mathematics	Programming	Elective	Elective
	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Elective	Elective
2028	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	STAT2000 Applied Statistics and Research Methods	STAT3800 Deterministic and Stochastic Optimisation
2029	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT2020 Predictive Analytics	STAT2300 Statistical Inference	STAT3040 Time Series Forecasting
	Semester 1	STAT3030 Generalised Lin- ear Models	Elective	Elective	
2030	Semester 2	STAT3100 Systems Think- ing for an Inte- grated Workforce			
	Semester 1				

2.3 Major: Studies in Mathematics and Statistics

Major course options

As part of this major you only need to choose two of the following 2000s level courses. This table separates the course by the semester they are offered to help plan.

Semester 1	Semester 2
MATH2800 - Ordinary Differential Equations and Applied Linear Algebra STAT2000 - Applied Statistics and Research Methods	MATH2242 - Complex Analysis STAT2300 - Statistical Inference

Additionally, you only need to choose four of the following 3000s level courses.

Semester 1	Semester 2
MATH3120 - Algebra MATH3205 - Applied Fourier Analysis MATH3820 - Numerical Methods STAT3030 - Generalised Linear Models STAT3800 - Deterministic and Stochastic Optimisation	MATH3170 - Number Theory MATH3700 - Partial Differential Equations STAT3040 - Time Series Forecasting STAT3100 - Systems Thinking for an Integrated Workforce

2.3.1 Semester 1 entry; starting with MATH1110

2027	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	Semester 1 2000s course or Elective	Semester 1 2000s course or Elective
	Semester 2	Elective	STAT2020 Predictive Analytics	Elective or Semester 2 2000s course	Elective or Semester 2 2000s course
2029	Semester 1	Semester 1 3000s course	Semester 1 3000s course or Elective	Semester 1 3000s course or Elective	Semester 1 3000s course or Elective
	Semester 2	MATH3600 Work Integrated Learning for Mathematical Sciences	Elective or Semester 2 3000s course	Elective or Semester 2 3000s course	Elective or Semester 2 3000s course

2.3.2 Semester 1 entry; starting with MATH1002

2027	Semester 1	MATH1002 Foundational Studies in Mathematics	STAT1100 Data Communication and Modelling	Programming	Elective
	Mid-year	MATH1110 Mathematics for Engineering, Science and Technology 1			
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	Semester 1 2000s course or Elective	Semester 1 2000s course or Elective
	Semester 2	STAT2020 Predictive Analytics		Elective or Semester 2 2000s course	Elective or Semester 2 2000s course
2029	Semester 1	Semester 1 3000s course	Semester 1 3000s course or Elective	Semester 1 3000s course or Elective	Semester 1 3000s course or Elective
	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	Elective or Semester 2 3000s course	Elective or Semester 2 3000s course	Elective or Semester 2 3000s course

2.3.3 Semester 1 entry; starting with MATH1002 (no midyear)

You cannot take both MATH2242 and MATH2800 and finish in 3 years. If you wish to only take STAT2000 and STAT2300, you will have an elective in the third year. We recommend you take MATH1110 during the 2027 Mid-year term, so that you can then continue with the previous pathway from 2027 Semester 2 onwards.

2027	Semester 1	MATH1002 Foundational Studies in Mathematics	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1300 Fundamentals of Statistics	Elective	Elective
2028	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT2000 or Elective	Elective	Elective
	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	Elective or STAT2300
2029	Semester 1	MATH2360 Linear Algebra	MATH2800 or Semester 1 3000s course	Semester 1 3000s course	Semester 1 3000s course
	Semester 2	MATH2370 Real Analysis	MATH3600 Work Integrated Learning for Math- ematical Sciences	Semester 2 3000s course or MATH2242	Semester 2 3000s course

2.3.4 Semester 2 entry; starting with MATH1110

Note: For the 2000s level courses, the MATH courses must be taken in 2029 to satisfy assumed knowledge.

2027	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1300 Fundamentals of Statistics	Programming	Elective
	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	Semester 1 2000s course or Elective	Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	Elective or Semester 2 2000s course
	Semester 1	MATH2360 Linear Algebra	Semester 1 2000s course or Elective	Semester 1 3000s course	Semester 1 3000s course or Elective
2029	Semester 2	MATH2370 Real Analysis	MATH3600 Work Integrated Learning for Math- ematical Sciences	Elective or Semester 2 2000s course	Elective or Semester 2 3000s course
	Semester 1	Semester 1 3000s course	Semester 1 3000s course	Elective	Elective
2030	Semester 1	Semester 1 3000s course	Semester 1 3000s course	Elective	Elective

2.3.5 Semester 2 entry; starting with MATH1002

2027	Semester 2	MATH1002 Foundational Studies in Mathematics	Programming	Elective	Elective
	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Elective	Elective
2028	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	Semester 1 2000s course or Elective	Semester 1 2000s course or Elective
2029	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT2020 Predictive Analytics	Elective or Semester 2 2000s course	Elective or Semester 2 2000s course
	Semester 1	Semester 1 3000s course	Semester 1 3000s course	Semester 1 3000s course	Semester 1 3000s course
2030	Semester 1	Semester 1 3000s course	Semester 1 3000s course	Semester 1 3000s course	Semester 1 3000s course

3 Bachelor of Mathematics (Advanced)

Majors

There are three (3) possible majors for the Bachelor of Mathematics:

- **Pure and Applied Mathematics:** For those who wish to immerse themselves in fundamental mathematics such as algebra and number theory, and to model real world scenarios by constructing and solving ordinary and partial differential equations by hand and on a computer.
- **Statistics:** For those that want to deal with the uncertainty of life, modelling real world systems with different data types, optimising models to better reflect data, and predicting the future with time series forecasting.
- **Studies in Mathematics and Statistics:** This major is for those who want to pick and choose courses from the other two majors.

Entry point

Many students will enter university after having completed their NSW HSC in the last two years. If you have successfully completed Extension 1 or 2 Mathematics or equivalent then you can skip MATH1110 and take MATH1120 immediately. **This enrolment must be done manually, as the system can't handle this case.** Contact Program Advice or the Program Convenor for this manual enrolment.

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3.1 Major: Pure and Applied Mathematics

MATH3000s course options

As part of this major you only need to choose four of the five MATH3000s level courses. The MATH3000s courses can be broadly split between pure and applied, use the following table to guide your choice of third year courses. All five MATH3000s level courses can be taken by using a generic elective slot.

	Skews pure	Skews applied
Semester 1	MATH3120 - Algebra MATH3205 - Applied Fourier Analysis	MATH3820 - Numerical Methods
Semester 2	MATH3170 - Number Theory	MATH3700 - Partial Differential Equations

3.1.1 Semester 1 entry; without MATH1110

2027	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH2310 Calculus of Science and Engineering	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	Elective	Elective
	Semester 2	MATH1510 Discrete Mathematics	STAT2020 Predictive Analytics	MATH2242 Complex Analysis	Elective
2029	Semester 1	MATH3101 Advanced Mathematical Topics I	Semester 1 MATH3000s course	Semester 1 MATH3000s course	SCIE3500 or Semester 1 MATH3000s course
	Semester 2	MATH3600 Work Integrated Learning for Mathematical Sciences	MATH3102 Advanced Mathematical Topics II	Semester 2 MATH3000s course	Semester 2 MATH3000s course or SCIE3500

Note: MATH3170, MATH3205, MATH3700, and MATH3820 can be taken in 2028 to free up an elective in 2029. MATH1510 and MATH2370 can be swapped.

3.1.2 Semester 1 entry; with MATH1110

2027	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	Elective
	Semester 2	MATH1510 Discrete Mathematics	STAT2020 Predictive Analytics	MATH2242 Complex Analysis	Elective
2029	Semester 1	MATH3101 Advanced Mathematical Topics I	Semester 1 MATH3000s course	Semester 1 MATH3000s course	Elective or Semester 1 MATH3000s course
	Semester 2	MATH3600 Work Integrated Learning for Mathematical Sciences	MATH3102 Advanced Mathematical Topics II	Semester 2 MATH3000s course	Semester 2 MATH3000s course or Elective

Note: MATH3170 and MATH3700 can be taken in 2028 semester 2. MATH1510 and MATH2370 can be swapped.

3.1.3 Semester 2 entry; without MATH1110

2027	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1510 Discrete Mathematics	STAT1300 Fundamentals of Statistics	Programming
	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	STAT1100 Data Communication and Modelling	Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT2020 Predictive Analytics	MATH2242 Complex Analysis
	Semester 1	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	Semester 1 MATH3000s course	Semester 1 MATH3000s course	Semester 1 MATH3000s course or Elective
2029	Semester 2	MATH3101 Advanced Math- ematical Topics I	MATH3600 Work Integrated Learning for Math- ematical Sciences	Semester 2 MATH3000s course	Elective or Semester 2 MATH3000s course
	Semester 1	MATH3102 Advanced Mathe- matical Topics II	SCIE3500 Research Inte- grated Learning	Elective	Elective

Note: MATH1510 and MATH2370 can be swapped.

3.1.4 Semester 2 entry; with MATH1110

2027	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	MATH1510 Discrete Mathematics	STAT1300 Fundamentals of Statistics	Programming
	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	Elective	Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	MATH2370 Real Analysis	STAT2020 Predictive Analytics
	Semester 1	MATH2360 Linear Algebra	MATH2800 Ordinary Differential Equations and Applied Linear Algebra	MATH3205 or MATH3820	Elective
2029	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	MATH2242 Complex Analysis	Semester 2 MATH3000s course	Elective
	Semester 1	MATH3101 Advanced Math- ematical Topics I	MATH3102 Advanced Mathe- matical Topics II	Semester 1 MATH3000s course	Semester 1 MATH3000s course
2030	Semester 2				
	Semester 1				

3.2 Major: Statistics

3.2.1 Semester 1 entry; without MATH1110

2027	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH2310 Calculus of Science and Engineering	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2360 Linear Algebra	STAT2000 Applied Statistics and Research Methods	Elective	Elective
	Semester 2	MATH1510 Discrete Mathematics	STAT2020 Predictive Analytics	STAT2300 Statistical Inference	Elective
2029	Semester 1	MATH3101 Advanced Mathematical Topics I	STAT3030 Generalised Linear Models	STAT3800 Deterministic and Stochastic Optimisation	SCIE3500 Research Integrated Learning
	Semester 2	MATH3600 Work Integrated Learning for Mathematical Sciences	MATH3102 Advanced Mathematical Topics II	STAT3040 Time Series Forecasting	STAT3100 Systems Thinking for an Integrated Workforce

Note: MATH1510 and MATH2370 can be swapped.

3.2.2 Semester 1 entry; with MATH1110

2027	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	STAT2000 Applied Statistics and Research Methods	Elective
	Semester 2	MATH1510 Discrete Mathematics	STAT2020 Predictive Analytics	STAT2300 Statistical Inference	Elective
2029	Semester 1	MATH3101 Advanced Mathematical Topics I	STAT3030 Generalised Linear Models	STAT3800 Deterministic and Stochastic Optimisation	Elective
	Semester 2	MATH3600 Work Integrated Learning for Mathematical Sciences	MATH3102 Advanced Mathematical Topics II	STAT3040 Time Series Forecasting	STAT3100 Systems Thinking for an Integrated Workforce

Note: MATH1510 and MATH2370 can be swapped.

3.2.3 Semester 2 entry; without MATH1110

2027	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1510 Discrete Mathematics	STAT1300 Fundamentals of Statistics	Programming
	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	STAT1100 Data Communication and Modelling	STAT2000 Applied Statistics and Research Methods
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT2020 Predictive Analytics	STAT2300 Statistical Inference
	Semester 1	STAT3030 Generalised Lin- ear Models	STAT3800 Deterministic and Stochastic Optimisation	Elective	Elective
2029	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT3040 Time Series Forecasting	STAT3100 Systems Think- ing for an Inte- grated Workforce	Elective
	Semester 1	MATH3101 Advanced Math- ematical Topics I	MATH3102 Advanced Mathe- matical Topics II	SCIE3500 Research Inte- grated Learning	Elective
2030	Semester 1	MATH3101 Advanced Math- ematical Topics I	MATH3102 Advanced Mathe- matical Topics II	SCIE3500 Research Inte- grated Learning	Elective

Note: MATH1510 and MATH2370 can be swapped.

3.2.4 Semester 2 entry; with MATH1110

2027	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	MATH1510 Discrete Mathematics	STAT1300 Fundamentals of Statistics	Programming
	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	STAT2000 Applied Statistics and Research Methods	Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	STAT2300 Statistical Inference
	Semester 1	MATH2360 Linear Algebra	STAT3030 Generalised Lin- ear Models	STAT3800 Deterministic and Stochastic Optimisation	Elective
2029	Semester 2	MATH2370 Real Analysis	MATH3600 Work Integrated Learning for Math- ematical Sciences	STAT3040 Time Series Forecasting	STAT3100 Systems Think- ing for an Inte- grated Workforce
	Semester 1	MATH3101 Advanced Math- ematical Topics I	MATH3102 Advanced Mathe- matical Topics II	Elective	Elective
2030	Semester 1				

3.3 Major: Studies in Mathematics and Statistics

Major course options

As part of this major you only need to choose two of the following 2000s level courses. This table separates the course by the semester they are offered to help plan.

Semester 1	Semester 2
MATH2800 - Ordinary Differential Equations and Applied Linear Algebra STAT2000 - Applied Statistics and Research Methods	MATH2242 - Complex Analysis STAT2300 - Statistical Inference

Additionally, you only need to choose four of the following 3000s level courses.

Semester 1	Semester 2
MATH3120 - Algebra MATH3205 - Applied Fourier Analysis MATH3820 - Numerical Methods STAT3030 - Generalised Linear Models STAT3800 - Deterministic and Stochastic Optimisation	MATH3170 - Number Theory MATH3700 - Partial Differential Equations STAT3040 - Time Series Forecasting STAT3100 - Systems Thinking for an Integrated Workforce

3.3.1 Semester 1 entry; without MATH1110

2027	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	MATH2370 Real Analysis	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2360 Linear Algebra	Semester 1 2000s course or Elective	Semester 1 2000s course or Elective	Elective
	Semester 2	MATH1510 Discrete Mathematics	STAT2020 Predictive Analytics	Elective or Semester 2 2000s course	Elective or Semester 2 2000s course
2029	Semester 1	MATH3101 Advanced Mathematical Topics I	SCIE3500 Research Integrated Learning	Semester 1 3000s course	Semester 1 3000s course
	Semester 2	MATH3102 Advanced Mathematical Topics II	MATH3600 Work Integrated Learning for Mathematical Sciences	Semester 2 3000s course	Semester 2 3000s course

Note: All 3000s level course except MATH3120 and STAT3030 can be taken in 2028 (with care that assumed knowledge is still met) to free up an elective in 2029. MATH1510 and MATH2370 can be swapped.

3.3.2 Semester 1 entry; with MATH1110

2027	Semester 1	MATH1110 Mathematics for Engineering, Science and Technology 1	STAT1100 Data Communication and Modelling	Programming	Elective
	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH2370 Real Analysis	MATH1800 Mathematical Modelling	STAT1300 Fundamentals of Statistics
2028	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	Semester 1 2000s course or Elective	Semester 1 2000s course or Elective
	Semester 2	MATH1510 Discrete Mathematics	STAT2020 Predictive Analytics	Elective or Semester 2 2000s course	Elective or Semester 2 2000s course
2029	Semester 1	MATH3101 Advanced Math- ematical Topics I	Semester 1 3000s course	Semester 1 3000s course	Semester 1 3000s course or Elective
	Semester 2	MATH3102 Advanced Mathe- matical Topics II	MATH3600 Work Integrated Learning for Math- ematical Sciences	Semester 2 3000s course	Elective or Semester 2 3000s course

Note: MATH1510 and MATH2370 can be swapped.

3.3.3 Semester 2 entry; without MATH1110

2027	Semester 2	MATH1120 Mathematics for Engineering, Science and Technology 2	MATH1510 Discrete Mathematics	STAT1300 Fundamentals of Statistics	Programming
	Semester 1	MATH2310 Calculus of Science and Engineering	MATH2360 Linear Algebra	STAT1100 Data Communication and Modelling	Semester 1 2000s course or Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2370 Real Analysis	STAT2020 Predictive Analytics	Elective or Semester 2 2000s course
	Semester 1	Semester 1 2000s course or Elective	Semester 1 3000s course	Semester 1 3000s course	Semester 1 3000s course or Elective
2029	Semester 2	MATH3600 Work Integrated Learning for Math- ematical Sciences	Elective or Semester 2 2000s course	Semester 2 3000s course	Elective or Semester 2 3000s course
	Semester 1	MATH3101 Advanced Math- ematical Topics I	MATH3102 Advanced Mathe- matical Topics II	SCIE3500 Research Inte- grated Learning	Elective
2030	Semester 1	MATH3101 Advanced Math- ematical Topics I	MATH3102 Advanced Mathe- matical Topics II	SCIE3500 Research Inte- grated Learning	Elective

Note: There is a lot of potential to change when the major courses are taken. MATH1510 and MATH2370 can be swapped.

3.3.4 Semester 2 entry; with MATH1110

Note: For the 2000s level courses, the MATH courses must be taken in 2029 to satisfy assumed knowledge.

2027	Semester 2	MATH1110 Mathematics for Engineering, Science and Technology 1	MATH1510 Discrete Mathematics	STAT1300 Fundamentals of Statistics	Programming
	Semester 1	MATH1120 Mathematics for Engineering, Science and Technology 2	STAT1100 Data Communication and Modelling	Semester 1 2000s course or Elective	Elective
2028	Semester 2	MATH1800 Mathematical Modelling	MATH2310 Calculus of Science and Engineering	STAT2020 Predictive Analytics	Elective or Semester 2 2000s course
	Semester 1	MATH2360 Linear Algebra	Semester 1 2000s course or Elective	Semester 1 3000s course	Semester 1 3000s course or Elective
2029	Semester 2	MATH2370 Real Analysis	MATH3600 Work Integrated Learning for Math- ematical Sciences	Elective or Semester 2 2000s course	Elective or Semester 2 3000s course
	Semester 1	MATH3101 Advanced Math- ematical Topics I	MATH3102 Advanced Mathe- matical Topics II	Semester 1 3000s course	Semester 1 3000s course
2030	Semester 1				

4 Course List

4.1 MATH courses

MATH1002 - Foundational Studies in Mathematics

Course offered: Sem1, Sem2

Course description: Many relationships in the real world can be modelled via a mathematical function of one or more variables. This course introduces common functions of one variable used in the sciences, equips students with the tools of differential and integral calculus to analyse the properties of such functions, and develops an understanding of the role of functions and calculus in real-world systems. This course is suitable for students with a background in high-school mathematics who have not studied, or who have not succeeded in, 2 unit or higher mathematics courses at HSC level. Students cannot count MATH1002 for credit if they have previously passed MATH1100, MATH1110 or MATH1210.

MATH1110 - Mathematics for Engineering, Science and Technology 1

Course offered: Sem1, Midyear, Sem2

Course description: This course covers the parts of calculus and algebra which are fundamental to all of mathematics and its applications. In algebra, students learn concepts and symbolic manipulation when calculating with large numbers of variables. In calculus, they learn concepts used when working with continuously changing variables. Both ways of thinking are essential in the mathematics met by students in the Sciences, Engineering and Commerce. These concepts will be further explored in MATH1120.

MATH1120 - Mathematics for Engineering, Science and Technology 2

Course offered: Sem1, Sem2

Course description: This course covers the mathematics necessary to perform calculations in, and create models for, the real world of Science and Engineering. Specifically, it will demonstrate how to do mathematics in a three-dimensional world. The course describes the fundamental ideas of calculus of functions of one and two variables, differential equations and linear algebra. It continues from MATH1110 to complete a first year of Mathematics suitable for Science and Engineering students, and others for whom Mathematics is a tool. Students who wish to proceed to further mathematics studies at second year level are recommended to complete MATH2340 after MATH1120.

MATH1510 - Discrete Mathematics

Course offered: Sem2

Course description: Discrete mathematics is the study of mathematical structures that are discrete, separated or distinct; in contrast with calculus which deals with continuous change. It is an important area of pure and applied mathematics, as well as providing the mathematical basis for the understanding of computers and modern computation. Discrete Mathematics is important in the sciences, where it has increasing application in many areas, an exemplar of which is the understanding of DNA sequences in molecular biology. The Discrete Mathematics course introduces first year students to the basic concepts of discrete mathematics, covering topics such as sets, logic, enumeration methods, probability, recurrence relations, induction and graph theory. The course provides important background for students pursuing a BMath degree. It covers much of the mathematics essential for students majoring in Computer Science or Software Engineering, and is a compulsory course in those degree programs.

MATH1800 - Mathematical Modelling

Course offered: Sem2

Course description: Mathematical modelling is key to bridging the gap between mathematics and solving real-world problems. Whether used to simulate the spread of an infectious disease or predict the trajectory of a falling satellite, mathematical modelling permeates disciplines across academia and industry. This course provides an overview of essential mathematical modelling techniques and their application to real-world problems, providing an introduction to the entire modelling process: model construction, analysis and research. Surveying general techniques, the course will also focus on a number of illustrative case studies covering topics across the spheres of the natural world and human endeavour, from environmental issues, to biology, ecology, medicine, nanotechnology, forensic sciences, engineering, business and commerce.

MATH2242 - Complex Analysis

Course offered: Sem2

Course description: Complex analysis forms a basis for not only advanced mathematical topics, including differential equations, number theory, operator theory and other 3000 and higher level courses, but also for special functions of mathematical and quantum physics. Through this, complex functions make a significant contribution to the understanding of the world in which we live. This course covers fundamental knowledge in the theory of analytical functions with applications to definite integration and culminates with study of harmonic and special functions.

MATH2310 - Calculus of Science and Engineering

Course offered: Sem1, Sem2

Course description: Provides the essential mathematical techniques of Physical Science and Engineering. These are the methods of Multivariable Calculus and Differential Equations. Multivariable Calculus involves a study of the differential and integral calculus of functions of two or more variables. In particular it covers introductory material on the differential calculus of scalar and vector fields, and the integral calculus of scalar and vector functions. Differential Equations arise from mathematical models of physical processes. Also includes the study of the main analytical and numerical methods for obtaining solutions to first and second order differential equations.

MATH2360 - Linear Algebra

Course offered: Sem1

Course description: Linear algebra is one of the central languages of science and engineering. Vectors and matrices lie at the heart of fields ranging from physics and engineering to computer graphics, signal processing, data science, and machine learning. Whenever large collections of data must be organised, transformed, or understood, linear algebra provides the essential tools. This course introduces the fundamental ideas of linear algebra and explores the subject as both a beautiful mathematical theory, grounded in precise definitions and rigorous proofs—and one of the most powerful and widely used subjects of mathematics today. Alongside the theory, we emphasise the geometric intuition behind key concepts and their practical use in applications such as data analysis, data processing, and machine learning. Through a blend of mathematical reasoning, computations in Python, and examples drawn from many disciplines, students will develop a deep understanding of the structure of linear algebra and the ability to apply these ideas to real-world problems.

MATH2370 - Real Analysis

Course offered: Sem2

Course description: A deeper understanding and experience in the formulation of well-reasoned mathematics is developed in this course. The focus of the course is elementary real analysis and notion of proof, which are fundamental for higher mathematics and its applications. Mathematical analysis emerges from the foundations of calculus, focusing on the rigorous study of limits involving functions, sequences, and series. This unit serves as an essential bridge between introductory calculus and advanced mathematical disciplines such as differential equations, topology, and Fourier analysis by establishing a firm theoretical framework and cultivating the ability to construct formal mathematical proofs. Designed for students of mathematics, science, and engineering, as well as future educators, the course explores why standard calculus practices work, providing the depth necessary for both high-level applications and clear classroom instruction. Beginning with the real number system, the course progresses through the limiting behaviour of sequences and series, leading to a rigorous study of functions of a single real variable and uniform convergence.

MATH2800 - Ordinary Differential Equations and Applied Linear Algebra

Course offered: Sem1

Course description: Differential equations provide one of the most powerful mathematical tools for understanding the natural world. Since rates of change are commonly expressed using derivatives, single and systems of differential equations arise whenever some continuously varying quantities and their rates of change in space or time are known or postulated. Whether seeking to understand biological or physical processes, behaviours of solids or liquids, ecological or mechanical systems, differential equations provide essential insights. If only one independent variable is involved, which is often time, these equations are called ordinary differential equations (ODEs). This course introduces students to the world of ODEs. The main focus of the course will be to investigate analytical and numerical solution techniques, qualitative behaviour of solutions and mathematical modelling to explore a wide breadth of application areas. A large component of the analysis of systems

of first order linear ODEs and of nonlinear systems near critical points involves applications of linear algebra techniques.

MATH3101 - Advanced Mathematical Topics I

Course offered: Sem1, Sem2

Course description: This course provides an in-depth study of a topic in mathematics at an advanced undergraduate level. The topic is individually chosen to suit the student's background and aspirations and will be offered either as a lecture course or as a reading course under the direction of a member of the academic staff.

The course content may be sourced from recognised external providers or professional bodies associated with the discipline.

Each student's topic must be approved by the program convenor.

MATH3102 - Advanced Mathematical Topics II

Course offered: Sem1, Sem2

Course description: This course provides an in-depth study of a topic in mathematics at an advanced undergraduate level. The topic is individually chosen to suit the student's background and aspirations and will be offered either as a lecture course or as a reading course under the direction of a member of the academic staff.

The course content may be sourced from recognised external providers or professional bodies associated with the discipline.

Each student's topic must be approved by the program convenor.

MATH3120 - Algebra

Course offered: Sem1

Course description: Extends the application of the familiar algebraic laws for adding and multiplying numbers, matrices and vectors to other contexts. Depending on just which laws are satisfied, the algebraic structures studied are called groups, rings and fields. These concepts underlie much of modern mathematics, and are essential background for research in any area of pure mathematics.

MATH3170 - Number Theory

Course offered: Sem2

Course description: Modern Society is built on a foundation of algorithms underpinned by the ubiquity of the computer, a discrete device that depends upon its programmed ability to understand the integers. In this course, we will use a foundation of paradigmatic algorithms to underpin our understanding of the integers, focussing on their properties and representations and how that informs our understanding and intuition of the mathematical world. This course provides an introduction to the important basic topics of number theory: prime numbers, factorisation, congruence and representation of numbers and Diophantine equations – via the use of fundamental algorithms.

MATH3205 - Applied Fourier Analysis

Course offered: Sem1

Course description: The course introduces Fourier analysis from an applied perspective. Fourier series and the Fourier transform are developed as links between the domains of time and frequency. Connections between the analogue and digital worlds are made through the sampling theorem and the Fast Fourier Transform. Applications to quantum physics are explored through Heisenberg uncertainty. Further applications to partial differential equations, tomography and probability/statistics are also explored.

MATH3600 - Work Integrated Learning for Mathematical Sciences

Course offered: Sem2

Course description: Work Integrated Learning (WIL) allows students to integrate real world skills and experience with theoretical concepts and knowledge obtained during their studies, improving graduate employability. Students will undertake a WIL experience relevant to those who have a developed knowledge of mathematics and/or statistics. Students may complete their WIL within a range of professional contexts including the private sector, government, university, NGOs or community organisations. The student will have an active role in

organising this WIL experience, however, students will need to consult with the Course Coordinator prior to approaching companies or organisations. The Course Coordinator will ensure the WIL experience is appropriate and does not adversely affect other existing WIL activities or courses of the University. A nominated supervisor within the host organisation will be required. In addition to the 140 hours of WIL, students will be expected to complete up to 40 hours of additional study load, which includes the completion of assessments, and on-campus contact hours.

MATH3700 - Partial Differential Equations

Course offered: Sem2

Course description: Partial differential equations arise from the mathematical modelling of a wide range of problems in biology, engineering, physical sciences, economics and finance. Therefore, they form an essential part of the mathematical background required for engineering and physical sciences. This course introduces students to the modern theory and methods of partial differential equations. It provides the students with the skills to formulate partial differential equations for modelling real-world problems, the knowledge to solve them using fundamental analytical and numerical methods, and the ability to interpret the results in the relation to the modelling context.

MATH3820 - Numerical Methods

Course offered: Sem1

Course description: Numerical methods are now at the heart of applied mathematics. Many significant practical problems cannot be solved by analytical methods - their solutions can only be approximated through numerical methods. Since numerical methods do not give exact solutions to problems it is important to analyse their accuracy. It is also important to understand the stability, efficiency and robustness of a numerical scheme. This course introduces concepts in numerical analysis emphasising the development of numerical algorithms to provide solutions to common problems formulated in science and engineering. This will develop the basic understanding of numerical algorithms, their computer implementation, applicability and limitations.

4.2 STAT courses

STAT1100 - Data Communication and Modelling

Course offered: Sem1

Course description: Extracting and presenting information from data lies at the heart of statistics and data science. In practice, data is not often in an immediately usable format, requiring management, manipulation and error-checking. Visualisation of data is an invaluable asset in discovering and presenting key information. This course provides a practical introduction to data wrangling and visualisation. Students will learn to tidy, validate, transform and summarise data, and design graphics that present insights from the data. Students will also develop coding skills using the Python language which is one of the most popular and preferred languages used in practice. No prior programming experience is necessary. Information about further statistics courses is available at the link: <https://www.newcastle.edu.au/school/information-physical-sciences/study/statistics-courses>

STAT1300 - Fundamentals of Statistics

Course offered: Sem2

Course description: Statistics provides us with a quantitative framework to utilise data for describing, summarising and modelling the world around us. This course provides students with fundamental concepts of probability, random variables and their distributions, and then applies them to provide the theoretical foundation for data analysis through statistical modelling, estimation and hypothesis testing. These concepts are motivated by real problems and illustrated with the leading statistical software platform R, a free, open-source, growing library of functions for object-oriented statistical programming. Both the theory and the introduction to R provide the foundation for subsequent 2000 and 3000 level statistics courses. The theory is also of general interest to users of statistical methods, reaching beyond the application of these methods to address their underlying principles, formal description and rigorous justification. The written assignments give you the opportunity to apply the concepts we learn in lectures to a number of problems, some theoretical and others computational involving the use of R. The topics covered will align with the course content covered by that stage of the semester. Interested in studying further statistics courses to develop your skills and improve your employability? Information about available statistics courses can be found here: <https://www.newcastle.edu.au/school/mathematical-and-physical-sciences/study/statistics-courses>

STAT2000 - Applied Statistics and Research Methods

Course offered: Sem1

Course description: This is an applied Statistics course which will meet the needs of practitioners in a wide range of science-related disciplines as well as budding statisticians. STAT2000 builds upon the basic techniques taught in the STAT1070 course. Practical data analysis is experienced including how to design appropriate research studies and collect data.

STAT2020 - Predictive Analytics

Course offered: Sem2

Course description: The world is awash in data and there is a huge demand for people with the skills and knowledge to turn data into actionable insights. STAT2020 covers the basics of predictive data analytics, statistical computing and visualisation. Students develop an understanding of data science, from the basic skills of data processing and visualisation to building sophisticated descriptive and predictive models. STAT2020 focuses on developing models for classification and prediction. The aim is to use a set of predictor variables to model the outcome of a target variable using techniques such as least squares and logistic regression, k-nearest neighbours, classification and regression trees (CART), and hierarchical classifiers. Shrinkage-based methods for model selection and cross-validation for model evaluation are also introduced. Students develop coding and reproducible reporting skills with open source software. STAT2020 equips students with the data skills and acumen to excel in their chosen career through their ability to analyse a variety of data sources and make data driven decisions.

STAT2300 - Statistical Inference

Course offered: Sem2

Course description: STAT2300 further extends the theory, methods and mechanics that underpin standard statistical analysis from STAT1300. The course builds upon advances in computational power by introducing exciting numerically-driven techniques such as jackknife and bootstrap resampling procedures for investigating and modelling data, thereby equipping students with tools for analysis of data when distributional assumptions are challenging. The exploration of these techniques will principally focus around aspects of estimation and hypothesis testing, including properties and relationships between estimators and tests. The methods developed in the course are demonstrated through examples using R, a leading, open-source statistical programming platform. The written assignments give you the opportunity to apply the concepts we learn in lectures to a number of problems, some theoretical and others computational involving the use of R. The topics covered will align with the course content covered by that stage of the semester.

STAT3030 - Generalised Linear Models

Course offered: Sem1

Course description: How do we model data of very different types in a consistent way? This course explores generalized linear models and illustrates how methods for analysing continuous and categorical data fit into this framework.

STAT3040 - Time Series Forecasting

Course offered: Sem1

Course description: This course provides a hands-on introduction to time series analysis, focusing on practical tools and techniques for analysing and forecasting time-dependent data. Using real-world examples from business, finance, science, and engineering, students will learn how to identify patterns in time series data, build appropriate forecasting models, evaluate their performance, and apply the results to support informed decision-making. The course is designed to be accessible to students from a range of disciplines, with an emphasis on applying concepts through the use of statistical software and data-driven projects.

STAT3100 - Systems Thinking for an Integrated Workforce

Course offered: Sem2

Course description: Through a quality improvement and data analytics lens, the course develops the student's ability to leverage workplace data and synergies, analyse and improve organisational systems, and realise quality processes and outcomes. Students will become systems thinkers, able to: engage appropriate statistical techniques for assessing, describing and improving systems, processes and outcomes; and, lead data-based decision making and quality improvement considerate of the system's many stakeholders, utilising the science,

psychology, and management of systems and the underlying statistical principles. The course provides a holistic industry-oriented approach which develops the student's analytical, management and leadership prowess and advances their T-shaped skillset. Without data you're just another person with an opinion.

STAT3800 - Deterministic and Stochastic Optimisation

Course offered: Sem1

Course description: From finance and health to science and engineering, optimisation has many applications and is at the heart of several modern technologies such as machine learning for big data and deep neural networks. This course develops the student's ability to understand and apply the fundamental analytical, computational and statistical techniques for optimising deterministic and stochastic problems in practice. The first part of the course deals with deterministic optimisation problems where all parameters are known, including linear and nonlinear programs. In practice, however, we often encounter systems, for which parameters are uncertain. The focus of the second part of the course is on methods for optimising stochastic systems, particularly where the dynamic of the system is governed by a Markov chain, such as in supply chains and queueing networks. The written assignments give students the opportunity to apply the concepts they learn in lectures and labs to a number of theoretical and computational problems. The topics align with the course content covered by that stage of the semester. The project gives students the opportunity to experience applying the concepts they learn in the course to a more applied problem as a teamwork. The project output involves a written report and verbal presentation.

4.3 Programming courses

ENGG1003 - Introduction to Engineering Programming with Python

Course offered: Sem1

Course description: This course introduces students to the fundamentals of procedural programming and computational problem-solving using the Python programming language, with a focus on engineering applications. Designed for students with no prior programming experience, the course builds foundational skills in writing, testing, and debugging code to solve real-world engineering problems.

INFT1004 - Introduction to Programming

Course offered: Sem1, Midyear, Sem2

Course description: This course introduces the key skills of problem solving and computer programming, including the elementary programming concepts of documentation, data elements, sequence, selection, and iteration.

SENG1110 - Object Oriented Programming

Course offered: Sem1, Midyear, Sem2

Course description: This course is an introduction to an object-oriented programming language. The course introduces the fundamentals of analysing a problem and then implementing a solution as a computer software system using an object-oriented language. Students learn about problem-solving strategies, top-down program development and programming style. The course provides a basic introduction to data abstraction and object-oriented analysis and design. Emphasis is placed on programming and testing.

4.4 Extra courses

SCIE3500 - Research Integrated Learning

Course offered: Sem1, Midyear, Sem2

Course description: This course provides extension material to encourage the development of advanced skills and enhance career progression. This is done through experiential learning supported by appropriate readings and career guidance that allows students to expand the theoretical knowledge and skills developed in their undergraduate program, and integrate these with real world skills and experience through undertaking a supervised practical project. Students will be able to undertake their project in a variety of settings (including but not limited to laboratory, field and desk-based investigations) and direction is provided on-site by a project supervisor. Assessment is based on a progress report, a research notebook, a final project report and an oral presentation. Students are expected to be highly self-motivated, proactive and maintain regular contact with their academic supervisor for project consultation and assistance. Admission to this course is based on merit and the number of places available may be restricted. Allocation to a project is not guaranteed and is

dependent on project availability and the suitability of applications for each project. Students should enquire through the Course Coordinator and/or Program Convenor for advice. To enrol in this course, students must obtain approval from the Course Coordinator in advance and request for a manual enrolment from the Course Coordinator. Note: Exemption from the GPA requirement will only be considered in special circumstances and each application for exemption will be considered on its own merit on a case-by-case basis. Students may make a case via written support from a nominated supervisor in their discipline area.

BMATH Courses

